Assignment - 04

Querying operations on relations # 2

1. List Faculty (ID, Name) for all courses of Autumn'2010.

Relational Algebra:

$$\pi_{instructorid,\ instructorname}(\sigma_{acadyear='2010' \wedge semester='Autumn'}(offers \bowtie_{offers.instructorid=instructor.instructorid} instructor))$$

Query:

```
SET SEARCH_PATH TO acad;
```

```
---1. List Faculty (ID, Name) for all courses of Autumn'2010.
SELECT DISTINCT i.instructorid, i.instructorname
FROM instructor i
JOIN offers o
    ON i.instructorid = o.instructorid
WHERE o.acadyear = '2010'
  AND o.semester = 'Autumn';
```

Output:

Data Output | Messages | Notifications

	instructorid [PK] character varying (5)	instructorname character varying (30)
1	MHR	Mehul Raval
2	PK	Pankaj Kumar

2. List Students (ID, Name) from batch 2008 registered for course MT101 in Autumn'2008.

Relational Algebra:

$$\pi_{studentid,name}(\sigma_{batch=2008 \wedge courseno='MT101' \wedge acadyear='2008' \wedge semester='Autumn'}(student \bowtie_{student.studentid\ =\ registers.studentid} registers))$$

Query:

```sql
---2. List Students (ID, Name) from batch 2008 registered for course MT101 in Autumn'2008.
SELECT DISTINCT s.studentid, s.name
FROM student s
JOIN registers r
    ON s.studentid = r.studentid
WHERE s.batch = 2008
  AND r.courseno = 'MT101'
  AND r.acadyear = '2008'
  AND r.semester = 'Autumn';
```

Output:

Data Output | Messages | Notifications

	studentid [PK] character varying (9)	name character varying (30)
1	200801001	Ram Kant
2	200801002	Akshya Singh
3	200801003	Shobha Gupta
4	200801004	Shikha Singh
5	200801005	Ashish Jain
6	200802001	Ram Kant
7	200802002	Akshya Singh
8	200802003	Shobha Gupta
9	200802004	Shikha Singh
10	200802005	Ashish Jain
11	200811001	Rama Kant
12	200811002	Akshya Gupta
13	200811003	Unnati Gupta
14	200811004	Mridula Singh
15	200811005	Amit Ajaad

3. List Students (ID, Name) from B Tech (progid='01') batch 2008 registered for course MT101 in Autumn'2008.

Relational Algebra:

$$\pi_{studentid,name}(\sigma_{progid='01' \wedge batch=2008 \wedge courseno='MT101' \wedge acadyear='2008' \wedge semester='Autumn'}(student \bowtie_{student.studentid=registers.studentid} registers))$$

Query:

```sql
---3. List Students (ID, Name) from B Tech (progid='01') batch 2008 registered for course MT101 in Autumn'2008.
SELECT DISTINCT s.studentid, s.name
FROM student s
JOIN registers r
    ON s.studentid = r.studentid
WHERE s.progid = '01'
  AND s.batch = 2008
  AND r.courseno = 'MT101'
  AND r.acadyear = '2008'
  AND r.semester = 'Autumn';
```

Output:

Data Output | Messages | Notifications

SQL

	studentid [PK] character varying (9)	name character varying (30)
1	200801001	Ram Kant
2	200801002	Akshya Singh
3	200801003	Shobha Gupta
4	200801004	Shikha Singh
5	200801005	Ashish Jain

4. List courses that were offered but students were not registered in those offerings.

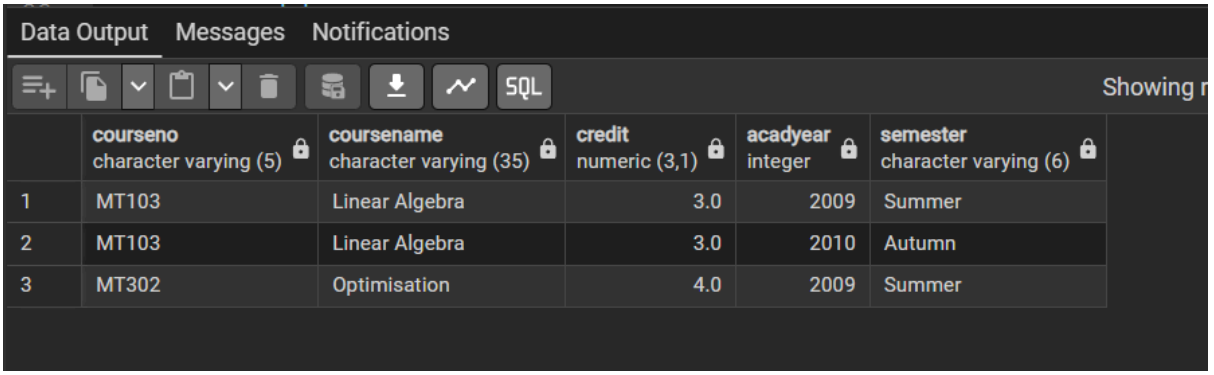
Relational Algebra:

$$course \bowtie (\pi_{acadyear,\ semester,\ courseno}(offers) - \pi_{acadyear,\ semester,\ courseno}(registers))$$

Query:

```sql
---4. List courses that were offered but students were not registered in those offerings.
SELECT DISTINCT
       o.courseno,
       c.coursename,
       c.credit,
       o.acadyear,
       o.semester
FROM offers o
JOIN course c
    ON o.courseno = c.courseno
LEFT JOIN registers r
    ON o.courseno = r.courseno
   AND o.acadyear = r.acadyear
   AND o.semester = r.semester
WHERE r.studentid IS NULL;
```

Output:



Data Output | Messages | Notifications

	courseno character varying (5)	coursename character varying (35)	credit numeric (3,1)	acadyear integer	semester character varying (6)
1	MT103	Linear Algebra	3.0	2009	Summer
2	MT103	Linear Algebra	3.0	2010	Autumn
3	MT302	Optimisation	4.0	2009	Summer

5. List students (id, name, courseno, grade) who got 'F' grade in Autumn'2008.

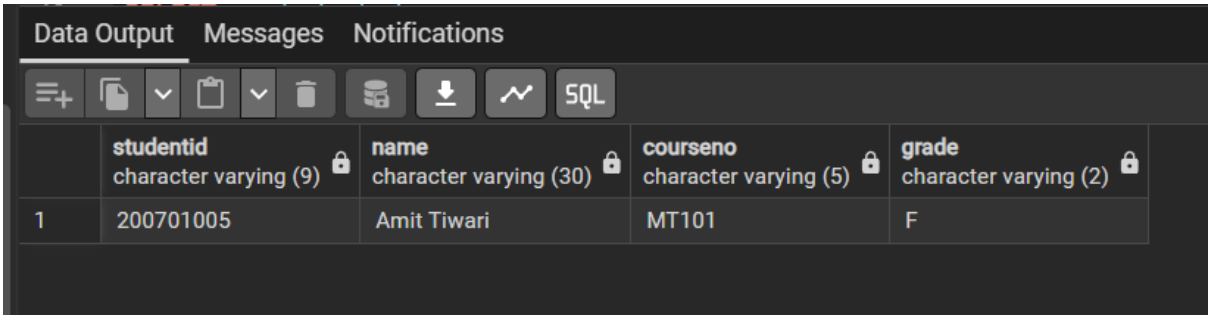
Relational Algebra:

$$\pi_{studentid,\, name,courseno,grade}(\sigma_{grade='F' \wedge acadyear='2008' \wedge semester='Autumn'}(student \bowtie_{student.studentid = registers.studentid} registers))$$

Query:

```
---5. List students (id, name, courseno, grade) who got 'F' grade in Autumn'2008
SELECT s.studentid,
       s.name,
       r.courseno,
       r.grade
FROM student s
JOIN registers r
    ON s.studentid = r.studentid
WHERE r.grade = 'F'
  AND r.acadyear = '2008'
  AND r.semester = 'Autumn';
```

Output:



Data Output | Messages | Notifications

	studentid character varying (9)	name character varying (30)	courseno character varying (5)	grade character varying (2)
1	200701005	Amit Tiwari	MT101	F

6. List Courses (Course No, Title, Credits) offered by Instructor (ID='PMJ') in semester Winter'2010.

Relational Algebra:

$$\pi_{courseno,coursename,credit}(\sigma_{instructorid='PMJ' \wedge acadyear='2010' \wedge semester='Winter'}(course \bowtie_{course.courseno=offers.courseno} offers))$$

Query:

```
---6. List Courses (Course No, Title, Credits) offered by Instructor (ID='PMJ') in semester Winter'2010.
SELECT DISTINCT c.courseno,
       c.coursename,
       c.credit
FROM course c
JOIN offers o
    ON c.courseno = o.courseno
WHERE o.instructorid = 'PMJ'
  AND o.acadyear = '2010'
  AND o.semester = 'Winter';
```

Output:

Data Output | Messages | Notifications

	courseno [PK] character varying (5)	coursename character varying (35)	credit numeric (3,1)
1	EC403	Settelite Communicati...	3.0
2	MT202	Adhoc Network	4.0

7. Produce transcript of student '200711002' for Academic year 2008-09 List Course No, Course Name, Course Credit, Grade.

Relational Algebra:

$$\pi_{courseno,coursename,credit,grade}(\sigma_{studentid\ =\ '200711002' \wedge acadyear='2008-09'}(registers \bowtie_{registers.courseno=course.courseno} course))$$

Query:

```
---7. Produce transcript of student '200711002' for Academic year 2008-09. List Course No, Course Name, Course Credit, Grade
SELECT c.courseno,
       c.coursename,
       c.credit,
       r.grade
FROM registers r
JOIN course c
    ON r.courseno = c.courseno
WHERE r.studentid = '200711002'
  AND r.acadyear = '2008';
```

Output:

Data Output | Messages | Notifications

	courseno character varying (5)	coursename character varying (35)	credit numeric (3,1)	grade character varying (2)
1	MT102	Computer Networks	4.0	AA
2	MT104	Probability	3.0	AA
3	MT105	VLSI	3.0	BB

Total rows: 3 | Query complete 00:00:00.154

8. List ID of students who have taken both courses - 'MT101' AND 'MT104'.

Relational Algebra:

$\pi_{studentid}(\sigma_{courseno='MT101'}(registers)) \cap \pi_{studentid}(\sigma_{courseno='MT104'}(registers))$

Query:

```
---8. List ID of students who have taken both courses - 'MT101' AND 'MT104'.
SELECT studentid
FROM registers
WHERE courseno IN ('MT101', 'MT104')
GROUP BY studentid
HAVING COUNT(DISTINCT courseno) = 2;
```

Output:

Data Output | Messages | Notifications

	studentid character varying (9)
1	200811001
2	200811002
3	200811003
4	200811004
5	200811005

9. List ID of students who have taken 'MT101' but not 'MT104'.

Relational Algebra:

$\pi_{studentid}(\sigma_{courseno='MT101'}(registers)) - \pi_{studentid}(\sigma_{courseno='MT104'}(registers))$

Query:

```sql
---9. List ID of students who have taken 'MT101' but not 'MT104'.
SELECT studentid
FROM registers
WHERE courseno = 'MT101'

EXCEPT

SELECT studentid
FROM registers
WHERE courseno = 'MT104';
```

Output:

Data Output | Messages | Notifications

	studentid character varying (9)
1	200702005
2	200701001
3	200702003
4	200801003
5	200802001
6	200701002
7	200701005
8	200702002
9	200801001
10	200801004
11	200802005
12	200801002
13	200802004
14	200702004
15	200701003
16	200702001
17	200802003
18	200801005
19	200701004
20	200802002